Question ID: a391ed22

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Equivalent expressions	Medium

Question

$$\left(\frac{1}{2}x+\frac{3}{2}\right)\left(\frac{3}{2}x+\frac{1}{2}\right)$$

The expression above is equivalent to ax^2+bx+c, where a, b, and c are constants. What is the value of b?

Rationale

The correct answer is $\frac{5}{2}$. The expression $\left(\frac{1}{2}x+\frac{3}{2}\right)\left(\frac{3}{2}x+\frac{1}{2}\right)$ can be written in the form ax^2+bx+c, where a, b, and c are constants, by multiplying out the expression using the distributive property of multiplication over addition. The result is $\left(\frac{1}{2}x\right)\left(\frac{3}{2}x\right)+\left(\frac{1}{2}x\right)\left(\frac{1}{2}\right)+\left(\frac{3}{2}\right)\left(\frac{3}{2}x\right)+\left(\frac{3}{2}\right)\left(\frac{1}{2}\right)$. This expression can be rewritten by multiplying as indicated to give $\frac{3}{4}x^2+\frac{1}{4}x+\frac{9}{4}x+\frac{3}{4}$, which can be simplified to $\frac{3}{4}x^2+\frac{10}{4}x+\frac{3}{4}$, or $\frac{3}{4}x^2+\frac{5}{2}x+\frac{3}{4}$. This is in the form ax^2+bx+c, where the value of b is $\frac{5}{2}$. Note that 5/2 and 2.5 are examples of ways to enter a correct answer.